

Credit Card Misuse Detection in NGAME

Can NGAME identify actual vs. intended usage of a company-issued credit card? Analysis of QBO ontology signals, detection strategies, and training period requirements.

Yes — NGAME can detect this, and it is well-suited to do so.

The detection gap between a transaction's **stated GL category** (how the bookkeeper coded it) and the **actual vendor type** (what the merchant actually sells) is precisely what NGAME's rich ontology extraction was designed to surface. QBO preserves enough signal — vendor name, account classification, memo, amount, date — to build a reliable actual-vs-intended usage detector.

5

Primary detection dimensions

12+

QBO fields with fraud signal

2

Core QBO CC transaction types

High

Vendor name signal strength

The Detection Gap: Intended vs. Actual

Asset misappropriation via credit card depends on a single exploitable gap: the perpetrator controls both the card and the expense report. The bookkeeper codes the transaction how they're told — but the **vendor name** is a ground-truth record that cannot be falsified without also falsifying the bank statement.

NGAME's ontology preserves the vendor name independently from the GL category assigned to it. A vendor classified under **Travel & Entertainment** that maps to a cruise line, auto dealer, or casino is a structural anomaly — regardless of memo text.

What the Card Issuer Intended	What the Card Was Actually Used For
Stated in policy; reflected in GL account coding	Preserved in vendor name — cannot be rewritten

Travel

Office Supplies

Meals & Entertainment

Fuel

Advertising

Professional Services

Casino / Gambling

Cruise Line

Auto Loan Payment

Personal Retailer

Subscription Service

Home Improvement

QBO Fields NGAME Preserves with Detection Value

All of these are available on Credit Card Charge and Credit Card Credit transaction types. NGAME's extraction retains them all as ontology properties.

QBO Field	What It Captures	Fraud Signal Type	Signal Strength
Vendor / Payee Name	The actual merchant (ground truth)	Actual-use classification	Critical
Account (GL Category)	How the bookkeeper coded the charge	Intended-use classification	Critical
Amount	Charge total	Structuring, threshold evasion, pattern	High
Date	Transaction date	Temporal patterns (weekends, holidays)	High
Payment Method	Which card / account	Card-purpose mapping	High
Memo / Description	Freeform note from user	Contradictory or missing justification	Medium
Class / Location	Program or site tag	Geographic / programmatic mismatch	Medium
Customer / Project	Billable assignment	Fictitious billing cover	Medium
Split line items	If charge was split across categories	Obscuring a single large personal charge	Medium

QBO Field	What It Captures	Fraud Signal Type	Signal Strength
Attachments present?	Whether a receipt was attached	Missing receipt = no audit trail	Medium
Tags	Custom labels	Policy tagging vs. actual use	Low–Medium
Reconciled flag	Whether matched to statement	Unreconciled personal charges left open	Low

Rows highlighted in green are the two highest-signal fields. The gap between them is the primary fraud detection mechanism.

Five Primary Detection Dimensions

1. Vendor-Category Mismatch (Highest Signal)

NGAME classifies each vendor name into its *actual* business type — independently of the GL account it was coded to. A charge to "Royal Caribbean" coded as "Travel" is technically travel-adjacent but should trigger a flag for personal luxury use, not business travel. A charge to "MGM Grand" coded as "Meals & Entertainment" is a casino.

Vendor Name Signal	Coded As	Actual Category	Risk
Royal Caribbean Intl	Travel	Cruise Line (personal leisure)	High
MGM Grand / Caesars	Meals & Entertainment	Casino / Gambling	Critical
Ally Financial / CarMax	Vehicle Expense	Personal Auto Loan / Purchase	Critical
Home Depot / Lowe's	Repairs & Maintenance	Home improvement (personal)	High
Netflix / Spotify / Hulu	Office Expense	Personal streaming subscription	High

Vendor Name Signal	Coded As	Actual Category	Risk
DraftKings / FanDuel	Advertising	Online gambling	Critical
Saks Fifth / Nordstrom	Uniforms / Clothing	Personal luxury retail	High
Spa / Massage / Salon	Professional Development	Personal wellness	Medium–High
Planet Fitness / Equinox	Employee Benefits	Personal gym membership	Medium

2. Temporal Pattern Anomalies

Legitimate business purchases cluster around business hours and workdays. Personal use bleeds onto weekends, late evenings, and holidays — especially for recurring personal subscriptions (same vendor, same day each month).

Weekend / Holiday Charges	Monthly Recurring Personal	Off-Cycle Spikes
Business supply cards charged Friday night through Sunday for non-travel vendors. Compare against employee PTO and travel calendar.	Same vendor, same amount, same day of month. Streaming services, gym memberships, auto loan payments — recurring personal billing is a fingerprint.	Large charges to personal-category vendors just before month-end close, before audits, or after approval chain changes — exploiting timing gaps.

3. Structuring / Threshold Evasion

Many small organizations have informal approval thresholds — charges below \$25, \$50, or \$100 may not require a receipt or manager approval. Perpetrators often split large personal purchases into multiple smaller charges, or use a vendor that processes naturally small amounts.

Pattern	Description	Detection Method
Clustering just below threshold	Multiple charges at \$24.xx, \$49.xx, \$99.xx to same vendor	Amount distribution clustering analysis

Pattern	Description	Detection Method
Same-vendor splits on same day	Two or more charges to same merchant on same day	Same-day duplicate vendor detection
Gradual escalation	Personal vendor charges growing slowly over months	Vendor spend trend analysis
Small round-number charges	\$20.00, \$50.00 to cash-like or ambiguous vendors	Round-amount flag on personal-category vendors

4. Card-Purpose Deviation

Many organizations issue cards with a stated purpose: one card for travel, one for supplies, one for fuel. NGAME can model the expected vendor-category distribution for each card and flag charges that fall outside that distribution.

In QBO this is representable through Payment Method tagging. If each card is a distinct payment method, NGAME can build a per-card vendor-category profile and flag drift — a fuel card used at a casino, or a supplies card at a cruise line.

5. Missing Audit Trail Indicators

In QBO, charges can carry attachments (receipts), memos, and reconciliation flags. Personal charges consistently lack receipts or carry vague memos ("supplies", "misc") — while charges to legitimate vendors tend to have documentation. This documentation gap is itself a signal.

Indicator	Personal Use Signal
No attachment on large charge	Deliberate — a cruise receipt or casino statement would be self-incriminating
Generic memo text	"Misc", "Supplies", "Expense" on vendors that should have specifics
Charge never reconciled	Perpetrator avoids drawing attention during bank reconciliation review
Charge split to obscure vendor	Line-item splits that bury the vendor name inside a multi-line entry

Why NGAME's Ontology Extraction Makes This Tractable

Standard GL analysis only sees the account code — it cannot distinguish a legitimate airline charge from a cruise line charge if both are coded "Travel." NGAME's extraction goes further, preserving the vendor name independently and enriching it with an inferred actual category.

What standard GL analysis sees

Account	Amount
6200 · Travel	\$3,847.00
6200 · Travel	\$412.00
6200 · Travel	\$89.00

All three look identical. Misuse is invisible.

What NGAME's ontology sees

Account	Vendor	Actual Type	Amount
Travel	Delta Airlines	Air Travel	\$3,847.00
Travel	Royal Caribbean	Cruise Line	\$412.00
Travel	Netflix	Streaming Sub	\$89.00

Vendor classification exposes the gap immediately.

Because NGAME preserves all transaction properties — not just the account code — it can enrich each transaction with an **inferred actual category** derived from vendor name analysis. The delta between coded category and inferred actual category is the primary misuse signal.

Training Period Requirements

Vendor classification is the foundational training task.

NGAME needs a ground-truth mapping of vendor names to actual business types before it can generate reliable misuse flags. This is the highest-leverage training investment.

Training Task	What It Requires	Output	Priority
Vendor → Actual-Type Taxonomy	Curated vendor names → business category labels (seed from MCC codes, Google/Yelp categories)	Vendor classifier or lookup table embedded in NGAME ontology	Critical
Per-Card Purpose Profile	Organization's stated card policies (which card is for what)	Expected vendor-category distribution per payment method	High
Baseline Spend Distribution	Historical charges by vendor-category, amount range, day-of-week	Statistical baseline for anomaly scoring	High
Known-Good Transaction Labels	Subset of transactions confirmed as legitimate (by auditor or management)	Positive training examples for classifier	High
Threshold / Approval Policy	Organization's receipt and approval thresholds	Structuring detection rules calibrated to org	Medium
Reconciliation Behavior Baseline	Historical pattern of which charge types get reconciled and when	Anomaly flags for unreconciled personal-type charges	Medium

Honest Limitations

NGAME operates on what QBO records. Several signals are absent or weak:

Limitation	Why It Matters	Potential Mitigation
No MCC codes in QBO	QBO does not import Merchant Category Codes from the card network — only the vendor name as the processor labeled it	NGAME builds its own vendor taxonomy from vendor names; can be augmented with bank feed enrichment if available

Limitation	Why It Matters	Potential Mitigation
Vendor name quality varies	Some card processors truncate or mangle names ("SQ *", "TST *", "AMZN MKTP")	Pattern matching and normalization layer; flag ambiguous vendor names for human review
No cardholder identity in transaction	QBO CC charges don't record which employee made the charge — only which card	Card-to-employee mapping must be maintained separately and joined at analysis time
Creative GL coding	A sophisticated perpetrator may reclassify charges to plausible categories (real vendor, wrong category)	Temporal and amount patterns still surface; vendor type is harder to fake than category